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# Wind load Analysis For Rolling Shutter

as per Indian Standards IS 875 Part III 1987

Project: SUSTEN
Client: MAHINDRA SUSTEN
Date: 08/11/2017
Height: 5000 mm
Width: 5000 mm
Slat Size: 75 mm
Slat Width: 15 mm

# Wind Load Analysis as per IS

Design Wind Speed Vz= Vb * k1 * k2 * k3

where Vb = Basic Wind Speed in m / s. = 25 m/s

k1 = Risk Coefficient (clause 5.) = 1

k2 = THS Factor (clause 5.3 = 1

k3 = Topography Factor (clause 5.) = 1

Vz = Design Wind Speed = 25 m/s

Terrain Category: 2

(Open terrain with well-scattered obstructions having heights between 1.5m to 10m)

Class of Building: A( Structures having max. dimensions is less then 20m)

k1, k2 & k3 constants are derived from IS-875, based on location height of structure & geographical factor. These factors are required for tall structures, Where we need to consider varying pressure condition.



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# Wind Load Analysis as per IS 875 - III

Design wind pressure = $0.6 \times V_z^2$

Where,

Pz = Design wind pressure

Vz = Design wind velocity

Wind Pressure, Pz = 0.6 x (25)2

Wind Pressure, Pz = 375 N/m^2

Pz = 0.000375 N/mm^2

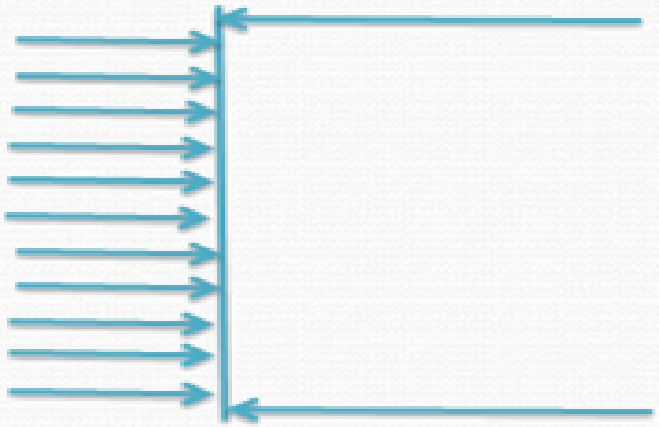


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Wind Load Analysis as per IS 875 - III

Assumed - Simply supported beam with uniform pressure.

375 N/m^2





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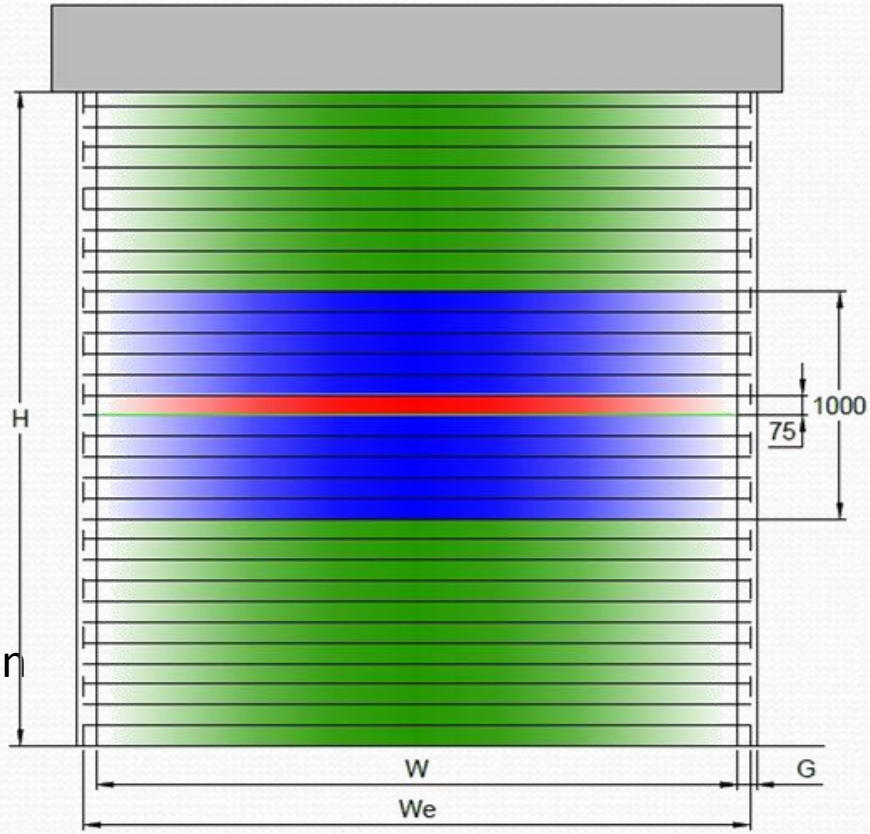
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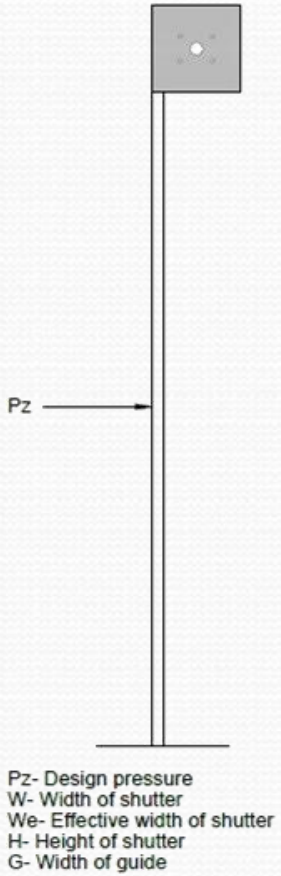
A)Bending Moment

B)Deflection

C)Maximum Stress

D)Side guides for curtain




Pz- Design pressure
W- Width of shutter
We- Effective width of shutter
H- Height of shutter
G- Width of guide



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## A)Bending Moment

Door is simply supported on two opposite sides

Max B.M. $M = \alpha \times W \times L^2/8$

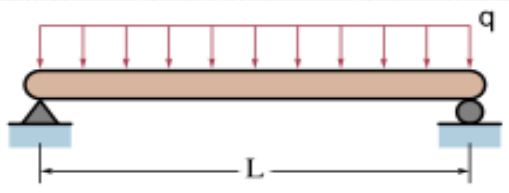
Where, W= UDL load

α= Bending moment coefficient as per IS 456:2000-Table26

## B)Deflection

Max. deflection will be at the centre of slat

D max = ( 5x M x L^2 ) / (48 x E x I)




Where ,

M = Bending Moment of Slat

E = Modules of Elasticity for CRS = 210GPa (KN/mm2)

I = Moment of Inertia of slat (From CAD Data)

L = Clear width of Shutter



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## C)Max Stress

Max. Stress= M x $\bar{y}$/ I

Where,

M=Bending moment of slat, $\bar{y}$= dist of side on y-y from CG , I= moment of inertia ,

Permissible Stress =248/FOS

=248/1.5

= 165 N/sq.mm

## D)Side Guides for Curtain panel

Side Guides required = w/60

Where,

w= width of shutter

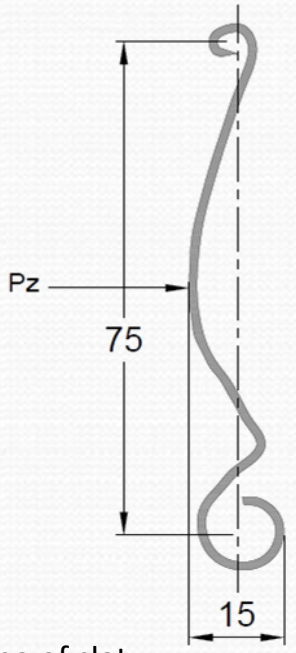
Design calculations for shutter of
Size: 5000mm(w) x 5000mm(H)

1. Wind Analysis for rolling shutter of size 5000mm(w) x 5000mm(h)

Design Wind Speed Vz= Vb * k1 * k2 * k3

where,

| | | |
|---|---|---|
| Vb = Basic Wind Speed in m / s. | = | 25 |
| k1 = Risk Coefficient (clause 5.4.1) | = | 1 |
| k2 = THS Factor (clause 5.3.2) | = | 1 |
| k3 = Topography Factor (clause 5.3.3) | = | 1 |
| Vz = Design wind Speed | = | 25 m/s |




Design wind pressure = 0.6x 25x25= 375 N/m2

UDL acting on Slat, W = Pz x A/L  Where, A= Area of slat

= 0.000375 x ( 5000 x 75 ) / 5000

W = 0.0281 N/mm

Bending Moment $M = \alpha.(W \times L^2)/8$

= 0.056 x 0.0281 x 5000 x 5000 / 8

= 4917.50 N-mm



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Ø Assuming thickness of slat 1.2 mm & minimum height as 75mm & depth as 15mm.

(From CAD DATA) I= 1564 mm^4

E= modulus of elasticity for CRS= 210 GPA

= 210000 N/mm^2

Max. deflection = 5 x 4917.5 x 5000 x 5000 / ( 48 x 210000 x 1564 )

= 38.99 mm

Permissible deflection = 5000/150

As per IS:800 = 33.333 mm

Max. stress = 4917.5x15/(2x1564)

= 23.58 N/mm^2

Permissible stress = (248 N/mm2) /1.5

= 165 N/mm^2

Guide side required = 5000/60 = 83.33

Hence we are using 100mm guide.

Note - As max. stress is less than permissible stress, hence no permanent deformation of slat will occur.

Thank You



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